Que 3 - Naive Bayes model using TF-IDF Vectors for Unigram+Bigram

		Training	**Testing**
Evaluate the Naive Bayes Model	Train Time	0.14 seconds	-
	F1 Score for Positive	0.90	0.90
	F1 Score for Neutral	0.15	0.09
	F1 Score for Negative	0.76	0.73
	AUC	0.9257	0.9091
	Accuracy	0.8270	0.8180
SMOTE + Class Weights on Naive Bayes Model	Train Time	0.32 seconds	-
	F1 Score for Positive	0.83	0.86
	F1 Score for Neutral	0.73	0.43
	F1 Score for Negative	0.80	0.70
	AUC	0.9203	0.8913
	Accuracy	0.7851	0.7546
Hyperparameter tuning	Train Time	3.66 seconds	-
	F1 Score for Positive	0.91	0.90
	F1 Score for Neutral	0.22	0.14
	F1 Score for Negative	0.76	0.73
	AUC	0.9294	0.9078
	Accuracy	0.8342	0.8212

Feature Engg - Name 2 features added (Categorical/Numerical Features)

- sentiment (Categorical - derived from mapping stars to Positive, Neutral, Negative).
- cleaned_text (Categorical - preprocessed version of text with tokenization, stopword removal, and lemmatization).

Interpretability Implemented

- Global: The analysis identifies key features influencing the model's predictions across all reviews.

2 Interesting Findings

- Words like "food", "service", and "place" influence all sentiments but carry different contexts (e.g., "poor service" vs. "excellent service").
- "Great" is highly indicative of Positive sentiment, showing its strong association with positive experiences.

Type of Cross Validation performed: 5-fold cross-validation (via cv=5 in GridSearchCV).

Findings of Cross Validation:
The best hyperparameter (alpha == 0.01) was selected based on the 5-fold cross-validation, optimizing model accuracy. The cross-validation helps prevent overfitting by ensuring the model is evaluated on different subsets of the data.

References:

1. https://chatgpt.com/share/6795b0ad-b418-8010-85d9-81b8c3131f90
2. https://chatgpt.com/share/6795b095-b288-8010-a827-3cbf5a0e8caf
3. https://chatgpt.com/share/6795b0dc-554c-8010-b4b4-dfa5077d0f10